

03-31-2025

Causal Inference w/ Unconfoundedness:

↳ unconfoundedness: $\swarrow$ binary treat

$$(Y(1), Y(0)) \perp\!\!\!\perp D \mid X$$

$$ATT = E[Y(1) - Y(0) \mid D=1]$$

$$= E[E[Y \mid X, D=1] - E[Y \mid X, D=0] \mid D=1]$$

↳ full steps in notes

by LIE: $E[Y \mid D=1] - E[E[Y \mid X, D=0] \mid D=1]$

$$ATE = E[E[Y \mid X, D=1] - E[Y \mid X, D=0]]$$

↳ Covariate Balance:

Suppose $D \perp\!\!\!\perp X$

$$ATT = E[Y \mid D=1] - E[E[Y \mid X, D=0] \mid D=0]$$

↳ only under covariate balance

$$= E[Y \mid D=1] - E[Y \mid D=0]$$

↳ Proxies: $(Y(1), Y(0)) \perp\!\!\!\perp D \mid (X, W)$

observed / unobserved

R - proxy variable (observed)

$$\textcircled{1} D \perp\!\!\!\perp (X, W) \mid R$$

$$\textcircled{2} (Y(1), Y(0)) \perp\!\!\!\perp R \mid (X, W)$$

↳ redundancy condition

$$ATT = E[Y \mid D=1] - E[Y(0) \mid D=1]$$

$$= E[Y \mid D=1] - E[E[Y(0) \mid R, D=1] \mid D=1] \text{ (L.I.E.)}$$

$$= E[Y \mid D=1] - E[E[E[Y(0) \mid X, W, R, D=1] \mid R, D=1] \mid D=1] \text{ (L.I.E.)}$$

$$= E[Y \mid D=1] - E[E[E[Y(0) \mid X, W, D=0] \mid R, D=1] \mid D=1]$$

↳ by redundancy + unconfoundedness

$$= E[Y \mid D=1] - E[E[E[Y(0) \mid X, W, R, D=0] \mid R, D=0] \mid D=1]$$

↳ by redundancy + unconfoundedness

↳ replace R; Flip D in 2nd E by ①

$$= E[Y \mid D=1] - E[E[Y \mid R, D=0] \mid D=1]$$

↳ by L.I.E (in reverse)

↳ got rid of W, unobserved var, allows for operation

